

CHAPTER 5

ADAPTATIONS TO ANAEROBIC TRAINING

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After completing this chapter, you will be able to

- differentiate between aerobic training adaptations and the anatomical, physiological, and performance adaptations following anaerobic training;
- discuss the central and peripheral neural adaptations to anaerobic training;
- understand how manipulating the acute training variables of a periodized program can alter bone, muscle structure, fiber types, and connective tissue;
- explain the acute and chronic effects of anaerobic training on the endocrine system;
- elucidate the chronic effects of anaerobic training on the cardiovascular system;
- understand the detrimental effects as a result of periods of detraining; and
- discuss how anaerobic training programs have the potential to enhance muscular strength, muscular endurance, power,

flexibility and mobility, and motor performance.

Characterized by high-intensity, intermittent bouts of exercise, **anaerobic training** requires adenosine triphosphate (ATP) to be regenerated at a faster rate than can be produced by oxidative metabolism alone. Consequently, the energy requirement relies on a greater proportion of energy coming from non-oxidative energy systems, which include the phosphagen or creatine phosphate system and a smaller proportion coming from glycolysis. Thus, the term *anaerobic training* is used in this text to describe exercise training that is performed at intensities that rely primarily on non-oxidative metabolism to meet energy demands. Example modalities include resistance training, plyometric drills, speed agility, and interval training. Long-term adaptations that occur in response to anaerobic training are specifically related to the characteristics of the training program. For example, improvements in muscular strength, power, hypertrophy, muscular endurance, motor skills, and coordination are all recognized as beneficial adaptations following anaerobic training modalities. Finally, while oxidative metabolism plays a limited role in regenerating ATP during high-intensity anaerobic activities, it still plays a critical role in the recovery of energy stores during periods of low-intensity exercise (recovery) or rest (49).

Exercises such as sprints and plyometric drills primarily stress the phosphagen system; they are usually less than 10 seconds in duration and minimize fatigue by allowing almost complete recovery between sets (e.g., 5-7 min). Longer-duration, high-intensity, interval-type training predominantly uses energy production from the glycolytic system, in which shorter rest intervals (e.g., 20-60 s) are adopted during high-intensity exercise. The integration of high-intensity exercise with short rest periods is considered an important aspect of anaerobic training, because athletes are often required to perform near-maximal intensities under fatigued conditions during competition. It is, however, critical that appropriate anaerobic

training be programmed and prescribed in such a way as to optimize the physiological adaptations that determine performance. Competitive sport requires the complex interaction of all the energy systems and demonstrates how each contributes to a varying extent to fulfill the global metabolic demands of competition ([table 5.1](#)).

A wide variety of physical and physiological adaptations are reported following anaerobic training, and these changes enable individuals to improve athletic performance standards ([table 5.2](#)). Adaptations include changes to the nervous, muscular, connective tissue, endocrine, and cardiovascular systems. They range from changes that take place in the early phase of training (e.g., one to four weeks) to those that take place following many years of consistent training. The majority of research has typically dealt with adaptations in the early to intermediate stages of training (i.e., 4-24 weeks). Understanding how the individual systems of the human body respond to physical activity using anaerobic metabolism provides a knowledge base from which the strength and conditioning professional can plan and predict the outcome of a specific training program to then focus on effectively influencing individual strengths and weaknesses.

TABLE 5.1 Primary Metabolic Demands of Various Sports

Sport	Phosphagen system	Glycolytic system	Aerobic system
American football	High	Moderate	Low
Archery	High	Low	—
Baseball	High	Low	—
Basketball	High	Moderate to high	Low
Boxing	High	High	Moderate
Cricket	High	Low	—
Diving	High	Low	—

Note: All types of metabolism are involved to some extent in all activities.

Sport	Phosphagen system	Glycolytic system	Aerobic system
Fencing	High	Moderate	—
Field events (athletics)	High	—	—
Field hockey	High	Moderate	Moderate
Golf	High	—	Moderate
Gymnastics	High	Moderate	—
Ice hockey	High	Moderate	Moderate
Lacrosse	High	Moderate	Moderate
Marathon running	Low	Low	High
Mixed martial arts	High	High	Moderate
Netball	Moderate	Moderate	High
Powerlifting	High	Low	—
Rowing	Low	Moderate	High
Rugby	High	High	Moderate
Skiing:			
Cross-country	Low	Low	High
Downhill	High	High	Moderate
Soccer (football)	High	Moderate	Moderate
Strongman	High	Moderate to high	Low
Swimming:			
Short distance	High	Moderate	—
Long distance	Low	Moderate	High
Tennis	High	Moderate	Low
Track (athletics):			
Sprints	High	Moderate	—
Middle distance	High	High	Moderate

Note: All types of metabolism are involved to some extent in all activities.

Sport	Phosphagen system	Glycolytic system	Aerobic system
Long distance	—	Moderate	High
Ultraendurance	—	—	High
Volleyball	High	Moderate	—
Water polo	High	High	High
Weightlifting	High	High	Moderate
Wrestling	High	Moderate	Low

Note: All types of metabolism are involved to some extent in all activities.

TABLE 5.2 Physiological Adaptations to Resistance Training

Variable	Resistance training adaptation
PERFORMANCE	
Muscular strength	Increases
Muscular endurance	Increases for high power output
Aerobic power	No change or increases slightly
Anaerobic power	Increases
Rate of force production	Increases
Vertical jump	Improved ability
Sprint speed	Improves
MUSCLE FIBERS	
Fiber cross-sectional area	Increases
Capillary density	No change or decreases
Mitochondrial density	Decreases
Myofibrillar density	No change
Myofibrillar volume	Increases
Cytoplasmic density	Increases

ATP = adenosine triphosphate; ATPase = adenosine triphosphatase.

Variable	Resistance training adaptation
Myosin heavy chain protein	Increases
Ribosomal capacity	No change or increases
Ribosomal efficiency	No change or increases
Myonuclear domain	Increases

ENZYME ACTIVITY

Creatine phosphokinase	Increases
Myokinase	Increases
Phosphofructokinase	Increases
Lactate dehydrogenase	No change or variable
Sodium–potassium ATPase	Increases

METABOLIC ENERGY STORES

Stored ATP	Increases
Stored creatine phosphate	Increases
Stored glycogen	Increases
Stored triglycerides	May increase

CONNECTIVE TISSUE

Ligament strength	May increase
Tendon strength	May increase
Collagen content	May increase
Bone density	No change or increases

BODY COMPOSITION

% body fat	Decreases
Fat-free mass	Increases

ATP = adenosine triphosphate; ATPase = adenosine triphosphatase.

Neural Adaptations

Many anaerobic training modalities emphasize the expression of muscular speed and power and depend greatly on optimal neural recruitment for maximal performance (and high quality of training). Anaerobic training has the potential to elicit long-term adaptations throughout the neuromuscular system, beginning in the higher brain centers and continuing down to the level of individual muscle fibers ([figure 5.1](#)). Neural adaptations are fundamental to optimizing athletic performance, and increased neural drive is critical to maximizing the expression of muscular strength and power. Augmented neural drive is thought to result in increased agonist (i.e., the major muscles involved in a specific movement or exercise) motor unit recruitment and potentially increased neuronal firing rates (3, 65, 140, 149). In addition, a reduction in inhibitory mechanisms (i.e., from Golgi tendon organs) and antagonist coactivation are also thought to occur with long-term training (3, 149). While it is not fully understood how these complex responses coexist, it is apparent that neural adaptations typically occur before any structural changes in skeletal muscle are apparent (167).

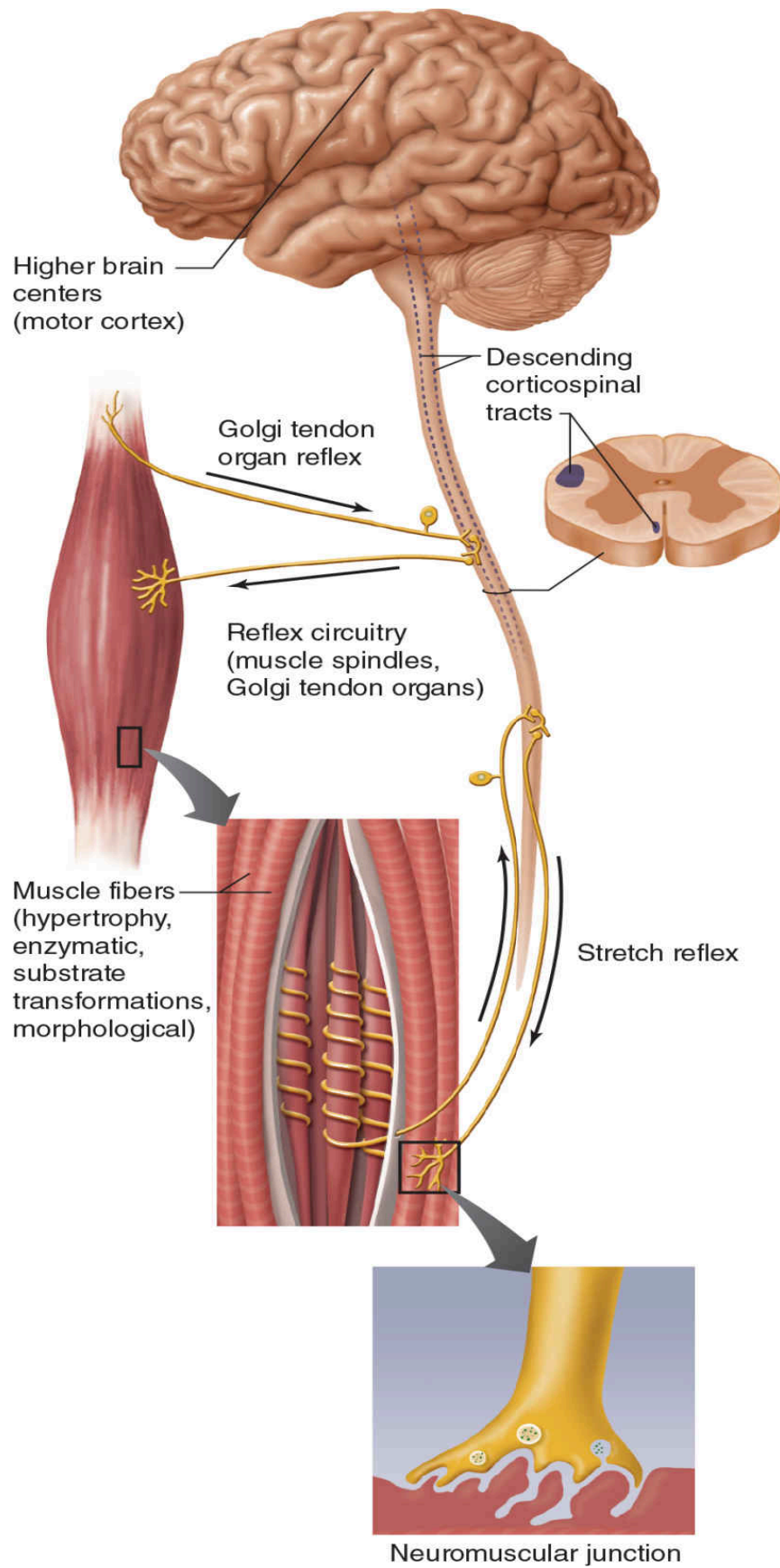


FIGURE 5.1 Potential sites of adaptation within the neuromuscular system.

Central Adaptations

Increased motor unit activation begins in the higher brain centers, where the intent to produce maximal levels of muscular force and power causes motor cortex activity to increase (45). As the level of force developed rises, or when a new exercise or movement is being learned, primary motor cortex activity is elevated in an effort to support the enhanced need for neuromuscular function. Adaptations to anaerobic training methods are then reflected by substantial neural changes in the spinal cord, particularly along the descending corticospinal tracts (4). Indeed, after use of anaerobic training programs, agonist voluntary activation and the recruitment of high-threshold motor units has been shown to be elevated in association with an improved skeletal muscle force production (130). This is in comparison to what is seen in untrained individuals (3), in whom the ability to maximally recruit motor units is limited. In those rehabilitating from injury, electrical stimulation has been shown to be more effective than voluntary contractions for eliciting strength gains (41). Other research has shown that only 71% of muscle tissue is activated during maximal efforts in untrained populations (7).

Adaptations of Motor Units

The functional unit of the neuromuscular system is the **motor unit**. Consisting of the alpha motor neuron and the muscle fibers that it activates, a motor unit may innervate <10 muscle fibers for small, intricate muscles or >100 fibers for large, powerful trunk and limb muscles. To maximize the expression of muscle force, large increases in synaptic input to the motor unit pool are necessary to cause the recruitment of all available motor units and to drive the discharge rate of these motor units to their optimal frequency. This association of motor unit recruitment and discharge rate modulation allows for the ability to finely grade muscle force production.

Therefore, changes in the firing rate (or discharge rate) of the motor unit also affect muscle force generation. With increased motor unit firing rates, the muscle fibers are continually activated by subsequent action potentials before the fibers have time to completely relax following a prior action potential. Therefore, increasing motor unit firing rates enhances the summation of successive muscle twitches, which is expressed as augmented contractile strength (1). Increases in motor unit firing rates represent a potential adaptive mechanism that correlates with improvements in strength in response to resistance training (166). Gains in maximal strength and power of agonist muscles are generally associated with an increase in recruitment; an increased rate of firing; greater motor unit synchronization of neural discharge intra-muscularly, and improved inter-muscle coordination, which is the learned motor behavior coordinating the activity of multiple muscles in synergy during movement (40a, 148); or a combination of all these factors.

The recruitment or decruitment of motor units in an orderly manner is governed by Henneman's **size principle** ([figure 5.2](#)), which represents the relationship between motor unit twitch force and recruitment threshold (140). According to this principle, motor units are recruited in an ascending order according to their soma size. Thus, motor units with small motor neurons have lower recruitment thresholds than do motor units with large motor neurons. High-threshold motor units are generally quick to fatigue but have greater force-producing ability. The recruitment threshold of a motor unit also determines its firing rate characteristics, and thus there is an inverse association between motor unit recruitment threshold and a motor unit's typical firing rate. This arrangement provides several advantages, including the ability to precisely gradate force production—particularly at low levels of force production—as well as enhanced fatigue resistance (29a, 38a).

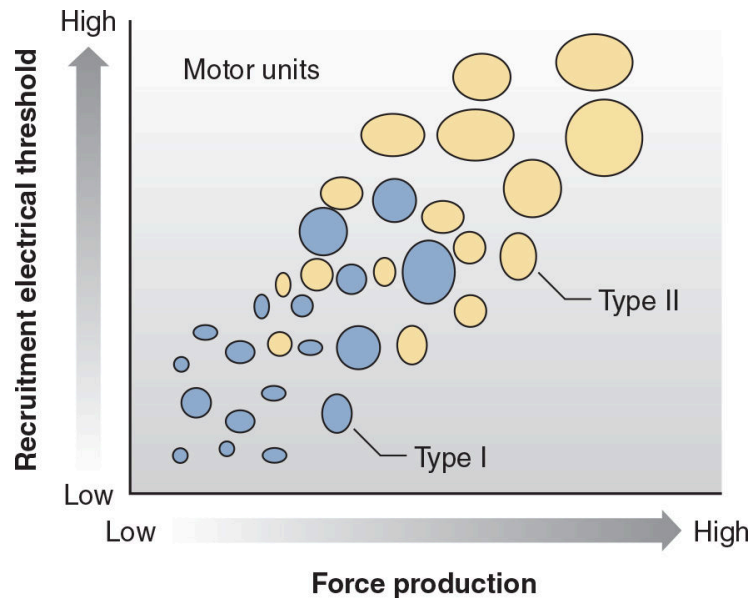


FIGURE 5.2 Graphic representation of the size principle, according to which motor units are organized based on the size of the neuronal soma. Low-threshold motor units are recruited first and have lower force capabilities than higher-threshold motor units. Typically, to get to the high-threshold motor units, the body must first recruit the lower-threshold motor units.

With heavy resistance training, all muscle fibers get larger (158, 159) because for the most part they are all recruited in order to produce the higher levels of force required to lift progressively heavier loads. It has similarly been shown that lifting lower loads near or to volitional failure can promote substantial hypertrophy of all fibers, because fatigue similarly demands progressive recruitment of fibers associated with higher-threshold motor units (121a, 122a). Maximal force production not only requires the recruitment of a maximum percentage of available motor units, including the high-threshold motor units, but also relies on the recruitment occurring at very high firing frequencies, which promotes summation of muscle fiber twitch forces and consequently augments the magnitude of contractile activity. Once a motor unit is recruited, less synaptic input is needed in order for it to be re-recruited (e.g., its recruitment threshold decreases) (65). This phenomenon may have important ramifications for strength and power training, because all motor

units, but especially high-threshold units, may be more readily reactivated subsequent to prior recruitment.

In addition to requiring the recruitment of a large number of available motor units, maximal force production also necessitates that these motor units fire at very high frequencies. This rapid firing promotes summation of the muscle fiber twitch forces, thereby amplifying the overall contractile activity. Studies have demonstrated that during maximal force contractions, motor neurons discharge at their highest rates, particularly in the early phases of the contraction when cortical input is dominant and afferent feedback from the muscle is minimal. This is especially evident during explosive, or ballistic, contractions when motor units are recruited in a precise, orderly sequence, with most units activated before a significant rise in force output. Consequently, even under conditions demanding rapid force, the recruitment of motor units adheres to an ordered pattern that prioritizes high-threshold motor units when maximum force is required. This orderly yet rapid recruitment pattern may be a key factor in developing strength and power, because it enables the reactivation of motor units, especially high-threshold ones, with reduced synaptic input after initial activation (40b).

With heavy resistance training, all muscle fibers get larger (i.e., hypertrophy) because more motor units are recruited to produce high levels of force.

Another critical element of adaptation in neural recruitment is the level of tissue activation that results from chronic resistance training for muscle hypertrophy (26a, 80a). Research has shown that as muscle hypertrophies, it does not require as much neural activation to lift a given submaximal load. Ploutz and colleagues (134) reported that fewer quadriceps muscle fibers were activated when subjects lifted a fixed submaximal load following nine weeks of resistance training, which also caused a 5% increase in muscle size. Additionally, following resistance training at high loads (80% 1RM),

the neural adaptation is greater, resulting in a lower neural output to produce the same submaximal force, even with similar hypertrophy to low-load (30% 1RM) resistance training (80a). Such results demonstrate the importance of progressive overloading during resistance training and how it promotes the activation of an optimal amount of muscle tissue to meet force production requirements.

Other motor unit adaptations include changes in the rate and sequence of firing. A positive relationship exists between the magnitude of force produced and the rate of motor unit firing; high firing rates from the onset of ballistic muscle contraction are especially critical to increased rates of force development (3). The increase in firing rate (vs. recruitment) appears to depend on muscle size and the number of motor units that compose the muscle, such that smaller muscles rely more on an increased firing rate to enhance force production, whereas larger muscles depend more on recruitment (61). Evidence suggests that anaerobic training can play a role in enhancing firing rates of recruited motor units (3). For example, resistance training may result in a more synchronized pattern (i.e., the firing of two or more motor units at a fixed interval) of activation during the exertion of large forces, rather than the customary asynchronous pattern usually common to motor function (149). Although the specific role of motor unit synchronization during anaerobic training remains to be fully elucidated, synchronization is potentially more critical to the rate of force development and less significant concerning the overall level of force developed.

Neuromuscular Junction

The **neuromuscular junction (NMJ)** is the interface between the nerve and the skeletal muscle fibers, and it represents another potential site for neural adaptation following anaerobic training (42, 43). Because of the difficulty in investigating this structure, most studies examining the NMJ have used animal models to demonstrate its adaptation to exercise. Deschenes and colleagues (44) examined

the impact of high-intensity versus low-intensity treadmill exercise training on NMJ in the soleus muscle of rats. Following both high- and low-intensity running, the total area of the NMJ was found to increase. High-intensity training, however, resulted in more dispersed, irregular-shaped synapses and a greater total length of nerve terminal branching compared to low-intensity training. In another study, greater end-plate perimeter length and area, as well as greater dispersion of acetylcholine receptors within the end-plate region, were also found after seven weeks of resistance training (43). These adaptations suggest that anaerobic training induces beneficial morphological changes in the NMJ that are conducive to enhanced neural transmission capabilities.

Neuromuscular Reflex Potentiation

Anaerobic training causes positive changes in the reflex (i.e., muscle spindle or stretch reflex) response of the neuromuscular system and enhances the magnitude and rate of force development via this reflex. This **myotatic reflex** also harnesses the involuntary elastic properties of the muscle and connective tissue (e.g., the series elastic component) and acts to positively increase force production. Resistance training in particular has been shown to increase reflex potentiation by between 19% and 55% (5). Furthermore, resistance-trained athletes (e.g., weightlifters and bodybuilders) are found to have greater reflex potentiation in the soleus muscle compared to untrained individuals (143).

Anaerobic Training and Electromyography Studies

Electromyography (EMG) is a common research tool used to examine the magnitude of neural activation within skeletal muscle. Two kinds of EMG are commonly used in research and applied settings: surface EMG and intramuscular (needle or fine wire) EMG. Surface EMG requires the placement of adhesive electrodes on the surface of the skin where they can monitor a large area of

underlying muscle (131). Surface EMG is often more effective for monitoring superficial muscle, because it is unable to detect deeper muscle activity. Also, the more body fat an individual has, the weaker the EMG signal will be because body fat acts as a low-pass filter. In comparison, with intramuscular EMG, the skin surface is numbed, and a needle electrode, or a needle containing two fine-wire electrodes, is inserted through the skin and positioned into the belly of the muscle itself. Fine-wire electrodes emphasize a specificity of assessment in that they are located in a muscle of interest and accurately record localized motor unit action potentials (75). Because of its invasiveness, intramuscular EMG is primarily adopted in research settings or under clinical conditions. While it is often difficult to determine the specific underpinning mechanism(s) (i.e., increased recruitment, discharge rate, or synchronization; Golgi tendon organ inhibition) affecting EMG output, an increase in EMG signal indicates greater neuromuscular activity.

An important consideration when examining the neuromuscular system is an individual's training status. Neural adaptations (improved motor learning and coordination) predominate in the early phase of training without substantial concomitant increases in muscle hypertrophy (61). It appears that as an individual's training status advances, an interplay exists between neural and hypertrophic mechanisms that contribute to further gains in strength and power.

Sale reported that dramatic increases in neural adaptation take place in the early part of a training program (6-10 weeks) (140). As the duration of training increases (>10 weeks), when muscle hypertrophy becomes more observable, it is these structural changes that contribute to strength and power gains more than neural adaptations (40c). Eventually, muscle hypertrophy plateaus as an accommodation to the training load occurs. However, at that time, if an athlete incorporates new variation or progressive overload into the training plan, neural adaptations will once again contribute to the performance improvements by acting to tolerate the new

physical insult from training. This pattern is replicated with every stepwise change in the training demand, and as athletes progress in training, the type of program used may be one of the most important factors to consider (72). Neural factors are especially important for strength gains in programs that use very high training intensities ($>80\%$ 1RM) (80a).

Electromyography studies have also yielded some interesting findings regarding neural adaptations to anaerobic training:

- Exercising muscle undergoing unilateral resistance training produces increased strength and neural activity in the contralateral resting muscle, a phenomenon known as **cross-education** (79). A literature review has shown that strength in the untrained limb may increase up to 22%, with an average strength increase of approximately 8% following training periods ranging from 4 to 12 weeks (125). The increase in strength of the untrained limb is accompanied by greater EMG activity in that limb (154), thereby suggesting that a central neural adaptation accounts for most of the strength gains, considering no change or reduction in the **cross-sectional area (CSA)** of muscle during the unilateral training period (22, 139).
- Unilateral strength, via cross-education, increases even when limbs are trained at different intensities and during detraining periods (22). The cross-education effect does appear to be load dependent, with untrained limb strength improving when the contralateral trained limb performs higher-load (8RM-12RM) intensities (22). Furthermore, the untrained limb strength improvements will also benefit if the untrained limb performs low-load (30RM-40RM) work while the trained limb performs high-load (8RM-12RM) synchronously, but not if the trained limb performs traditional unilateral work under low-load intensities (30RM-40RM) (22).
- In untrained individuals, a **bilateral deficit** is evident. The force produced when both limbs contract together is lower

than the sum of the forces they produce when contracting unilaterally. Research has shown that the corresponding EMG activity is lower during bilateral contractions (61), suggesting that neural mechanisms are, at least in part, a contributing factor. With longitudinal bilateral training, the magnitude of the bilateral deficit is reduced. In fact, trained or stronger individuals often show a **bilateral facilitation** effect whereby an increase in voluntary activation of the agonist muscle groups is observed during bilateral contraction (64a).

The EMG activity of antagonist muscle groups has been shown to change in response to anaerobic training during agonist movements. In most instances, cocontraction of antagonist muscles serves as a protective mechanism to increase joint stability and reduce the risk of injury (96). However, when too much antagonist activity opposes agonist movement, it inappropriately diminishes net force production at a joint.

A number of studies have shown reduced antagonist coactivation following resistance training, resulting in an increase in net force without an increase in agonist motor unit recruitment (71, 130). Sprint and plyometric training have also been shown to alter the timing of coactivation (86), although the specific role of altering antagonist coactivation patterns remains unclear. Greater antagonist activity may be observed during ballistic movements that require high levels of joint stability or when people are unfamiliar with a task and require more inherent stability (48). Antagonist activity has been reported to increase during unstable exercise activity with subsequent reduction in agonist activity (19, 144). Behm and colleagues (19) demonstrated that completing leg extension while seated on a Swiss ball (i.e., an unstable surface) resulted in 44% lower quadriceps activation but 29% greater hamstring activation when compared to a stable condition. Thus, leg extension forces were 70% lower in unstable than stable conditions.

Similarly, Santana and colleagues (144) demonstrated that pressing in a stable (bench press) versus unstable (standing cable

press) position requires an increase in antagonist, synergist, and core stabilization activation during the unstable condition. In contrast, agonist activation is greater during the stable condition with little effect on antagonist activation. The changes in agonist and antagonist activation result in the attenuation of force production. It has been shown that strength (1RM) performance is decreased in an unstable free-weight squat compared to a stable free-weight squat (116). Similarly, peak force (N) and rate of force development (RFD) have also been found to be significantly reduced during unstable squatting versus stable squatting (115). Additionally, when considering the type of instability apparatus, individual training status may be an important consideration. For example, among highly resistance-trained individuals, apparatuses that create moderate instability (e.g., Swiss ball, Dyna disc, and wobble board) did not substantially alter muscle activation. It is possible that this finding could be due to the highly trained individuals' familiarity and experience with unstable training conditions, but this is a hypothesis that must be tested (171).

Muscular Adaptations

Skeletal muscle adaptations following anaerobic training occur in both structure and function, with reported changes encompassing increases in fiber size, fiber type transitions, and enhanced biochemical and ultrastructural components (i.e., muscle architecture, enzyme activity, and substrate concentrations). Collectively, these adaptations result in enhanced performance characteristics that include strength, power, and muscular endurance, all of which are critical to athletic success.

Muscular Growth

Muscle **hypertrophy** is the term given to the enlargement of muscle fiber cross-sectional area (CSA) following training. A positive relationship exists between hypertrophy and the expression of muscular strength (162). Biologically, the process of hypertrophy involves an increase in the net accretion (i.e., an increase in synthesis, reduction in degradation, or both) of the contractile proteins **actin** and **myosin** within the myofibril, as well as an increase in the number of myofibrils within a muscle fiber. In addition to these contractile proteins, other structural proteins such as **titin** and **nebulin** are also synthesized proportionately to the myofilament changes. The new myofilaments are added to the periphery of the myofibril and result in an increase in its diameter. The cumulative effect of these additions is an enlargement of the fiber and, collectively, the size of the muscle or muscle group itself. During exposure to mechanical loading (e.g., resistance training), a series of intracellular processes regulate gene expression and subsequently promote increased protein synthesis (138).

Mechanical deformation of muscle stimulates various proteins independent of hormone concentrations, and these proteins have been shown to increase in activity before evidence of muscle hypertrophy appears. In particular, mechanical tissue deformation activates the protein kinase B (Akt)–mammalian target of rapamycin

(mTOR) pathway, the adenosine monophosphate–activated protein kinase (AMPK) pathway, and the mitogen-activated protein kinase (MAPK) pathway. Of these, the Akt/mTOR pathway in particular is important in directly regulating adaptations to resistance training (18, 45). When muscle fibers contract, Akt/mTOR signaling increases dramatically, and this response is critical for increasing muscle protein synthesis and subsequent growth (a process known as **myogenesis**). At the same time, the downregulation of inhibitory growth factors (e.g., myostatin) suggests that resistance exercise significantly affects a plethora of growth-signaling and breakdown pathways (23, 87, 89, 145). Protein synthetic rates are elevated after acute resistance exercise and remain elevated for up to 48 hours (109, 133). The magnitude of increased protein synthesis depends on a variety of factors including whether someone is trained or untrained, carbohydrate and protein intake, amino acid availability, timing of nutrient intake, mechanical stress of the workout, muscle cell hydration levels, and the anabolic hormonal and subsequent receptor response (14, 36, 91).

The process of hypertrophy involves both an increase in the synthesis of the contractile proteins actin and myosin within the myofibril and an increase in the number of myofibrils within the muscle fiber itself. The new myofilaments are added to the external layers of the myofibril, increasing its diameter.

Exercise-induced muscle damage (EIMD) and disruption of myofibrils and the uniform structure of muscle fiber sarcomeres following high-intensity anaerobic training (e.g., resistance training) also have a marked effect on muscle growth (35). The initial resistance exercise-induced stress in untrained individuals disrupts muscle homeostasis, leading to morphological changes such as cytoskeletal disturbances (e.g., Z-band streaming); loss of sarcomeric structural proteins (e.g., desmin and dystrophin);

segmental necrosis of muscle fibers; and alterations in connective tissue, T-tubules, and the sarcoplasmic reticulum. These morphological changes can result directly from mechanical stress during exercise or subsequently from calcium-sensitive degradation pathways and inflammatory responses. EIMD damage is most pronounced after the first resistance training session and diminishes rapidly as training continues, becoming negligible after 10 weeks of consistent resistance training (27, 37). The structural changes associated with EIMD influence gene expression in an effort to strengthen muscle tissue and protect it from further damage. The repair and remodeling process itself may involve a host of regulatory mechanisms (e.g., hormonal, immune, and metabolic) that interact with the training status of the individual (90). However, it is understood that both the inflammatory responses and increased protein turnover (i.e., increased net protein synthesis) ultimately contribute to long-term hypertrophic adaptations. The sequence of protein synthesis involves (a) water uptake, (b) noncontractile protein synthesis, and (c) contractile protein synthesis (166). At the same time, reduced degradation acts to maintain the size of fibers by reducing net protein loss.

After the initiation of a heavy resistance training program, changes in the type of muscle proteins (e.g., fast myosin heavy chains) start to take place within several workouts (158). However, muscle fiber hypertrophy requires a longer period of training (3-4 weeks of 9-12 workouts) before significant changes in CSA become detectable (40, 150). The hypertrophic process leading to a rise in CSA will have been occurring molecularly since the initial training stimulus, but is only observable when those alterations become measurable. As with initial gains in performance variables (e.g., strength and power), hypertrophic responses are at their greatest in the early stages, after which the rate of muscle growth diminishes over time (72). Athletes studied across two years of heavy resistance training showed increases in strength that paralleled optimal training

intensities, although muscle fiber hypertrophy contributed little to increased lifting performance (72).

The magnitude of hypertrophy ultimately depends on the training stimulus and how the acute training variables are prescribed. To optimize muscle growth, appropriate training periodization is essential for maximizing the combination of mechanical and metabolic stimuli. Mechanical factors include the lifting of heavy loads or low to moderate loads at near-momentary muscular failure, with or without the inclusion of eccentric muscle actions (159a), and moderate to high training volumes (95), all of which are characteristic of typical resistance training prescriptions for muscle strength and hypertrophy. An increasing amount of evidence also supports the use of novel training modalities (e.g., occlusion training) as alternative methods of inducing appropriate mechanical or metabolic stress. Metabolic factors center on low- to moderate-intensity or moderately high-intensity training with high volumes and short rest intervals (characteristic of bodybuilding training) (39, 136). Collectively, mechanical factors result in optimal recruitment of muscle fibers (because muscle fibers need to be recruited before growth), growth factor expression, and potential disruption to the sarcomeres, all of which increase muscle CSA (63). The metabolic factors stress the glycolytic energy system and result in increased metabolites that may be involved in muscle growth. These programs also elicit the most potent anabolic hormone response (96).

Also proposed as a mechanism for increasing muscle size, and something that has been debated by exercise scientists for years, is hyperplasia. **Hyperplasia** is the term given to an increase in the number of muscle fibers via longitudinal fiber splitting in response to high-intensity resistance training. Hyperplasia has been shown to occur in animals (13, 64, 77), but the findings are controversial in humans, with some studies supporting (108, 111, 114, 165) and others rejecting its occurrence in humans (110). Part of the reason for this confusion may arise from cross-sectional studies comparing resistance-trained athletes to untrained individuals, showing a

greater number of fibers in trained populations. In this scenario it is difficult to determine if the difference is due to preexisting, pretraining differences in fiber number (i.e., due to genetics) or due to training-induced hyperplasia. One must also consider that the procedures scientists perform on animals simply cannot be completed in humans for ethical and logistical reasons. Consequently, some of the more convincing data supporting skeletal muscle hyperplasia emerge from animal studies (64, 77). When examining hyperplasia in animal models, researchers remove whole muscle and strip back the fascia in order to analyze muscle fiber numbers longitudinally under a microscope. In humans, removal of whole muscle is not feasible. Instead, needle biopsy techniques are used to harvest a small sample of muscle tissue, which is then examined in cross section, and fiber number calculations are performed indirectly using extrapolation mathematics. While hyperplasia cannot be completely ruled out, it does not appear to be a major strategy for muscle tissue adaptation to resistance training, and, if it occurs at all, it involves only a small amount of the added tissue (maybe less than 10%) if the conditions are optimal. It may be hypothesized that if hyperplasia occurs, it may be in response to muscle fibers reaching a theoretical upper limit in size, which may occur in athletes using anabolic steroids and other growth agents or undergoing long-term training that produces large to extreme levels of hypertrophy. This remains to be fully elucidated and continues to be an area of contention among scientific researchers.

Fiber Type—Specific Size Changes

The magnitude of muscle hypertrophy experienced following anaerobic training is associated with muscle fiber type. Muscle fibers must be activated in order to promote significant hypertrophy (38). During resistance training, both type I and type II muscle fibers have the potential to be recruited, and only via activation do type I or type II fibers receive a signaling mechanism that initiates the cascade of regulatory processes promoting protein synthesis.

Because there is an apparent alignment of the functional characteristics of fiber types (e.g., fiber size, force capacity, and fatigue resistance) and motor unit properties, such that in mixed muscles low-threshold motoneurons typically innervate type I fibers and high-threshold motoneurons innervate type II fibers, it is necessary that high-force and/or high-fatigue conditions are used to maximize phenotypic adaptations across all fiber types. Accordingly, significant and consistent activation via a consistent training stimulus at appropriate volume is necessary for the initiation of these adaptive processes to translate to significant hypertrophy as observed by enlarged CSA of the intact muscle. Typically, type II fibers manifest greater increases in size than type I fibers; thus the magnitude of hypertrophy is not uniform between the two major fiber types (68). In fact, it has been argued that the ultimate potential for hypertrophy may reside in the relative proportion of type II fibers within a given athlete's muscles (165). That is, athletes who possess a relatively large proportion of fast-twitch fibers may have a greater potential for increasing muscle mass than individuals possessing predominantly slow-twitch fibers.

Fiber Type Transitions

The pattern of neural stimulation dictates the extent to which fiber type adaptations occur following anaerobic training. Muscle fibers are theoretically positioned on a continuum from the least oxidative to the most oxidative type (see [figure 5.3](#) and the description of muscle fiber types in [chapter 1](#)). The continuum is as follows: IIx, IIax, IIa, IIac, IIc, Ic, and I, with a concomitant myosin heavy chain (MHC) expression (i.e., MHC IIx, IIa, and I) (11, 157). Although the proportions of type I and II fibers are partially genetically determined (132, 154a), changes within each subtype can occur following anaerobic training. With training and activation of the high-threshold motor units, a transition from type IIx to IIa fiber type occurs (26). In other words, type IIx muscle fibers change their myosin adenosine triphosphatase (ATPase) isoform content and

progressively become more oxidative IIa fibers. In fact, research has shown nearly full transitions from type IIX to IIa fiber profiles following the combination of high-intensity resistance and aerobic endurance training (94).

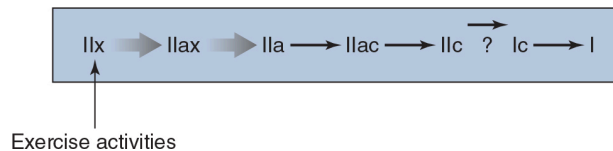


FIGURE 5.3 Muscle fiber transitions occur during training, and these changes are not influenced by sex (17, 158). This means that a shift of the type of myosin adenosine triphosphatase (ATPase) and heavy chains takes place during training. Transformations from IIX to IIax to IIa can be seen, and then small percentages change to IIac and IIc. Exercise activities that recruit motor units with type IIX muscle fibers initiate a shift toward IIa fibers.

On consistent activation, type IIX fibers change into a more oxidative form along the continuum (i.e., to an intermediate fiber type IIax and then to a IIa) (26). Any change in the muscle fiber type continuum and associated MHCs occurs in the early stages of a resistance training program. In an early study, Staron and colleagues (158) examined the effects of a high-intensity resistance training protocol (multiple sets of the squat, leg press, and knee extension exercises using 6RM to 12RM loads and 2-min rest periods) performed by men and women two times per week for eight weeks. They reported a significant decrease in the type IIX percentage in women after just two weeks of training (four workouts) and in the men after four weeks of training (eight workouts). Over the eight-week training program, type IIX fiber types decreased from approximately 18% to about 7% of the total muscle fibers in both men and women. Analysis of the MHCs showed that in this early phase of training, the proportion of IIX MHCs shifted toward IIa MHCs. Interestingly, detraining has the opposite effect, resulting in an increase in type IIX fibers and a reduction in type IIa fibers (132), with a possible overshoot of type IIX fibers (i.e., higher IIX

percentages than observed pretraining [11]). While transformation within muscle fiber subtypes appears typical, transformation from type I to type II or vice versa appears less probable, most likely due to differing MHC isoforms and relative oxidative enzyme content (132). Whether these transformations are possible remains doubtful, because insufficient evidence is currently available. These possibilities need to be explored in future studies; it is intriguing to consider whether the continuum of adaptations extends beyond type I and II subpopulations, especially in extreme cases in which a marathon runner (high percentage of type I fibers) follows a high-intensity resistance training program or a powerlifter (high percentage of type II fibers) begins an extensive aerobic endurance training program.

Structural and Architectural Changes

Pennate muscle has fascicles that attach obliquely (in a slanted position) to its tendon. **Pennation angle** affects the force production capabilities as well as the range of motion of a muscle. Larger pennation angles can also accommodate greater protein deposition and allow for greater increases in CSA (2). In pennate muscle, resistance training has been shown to increase the angle of pennation, with strength-trained athletes displaying larger pennation angles in the triceps brachii and vastus lateralis muscles compared to untrained individuals (2). Additionally, fascicle length has been shown to be greater in strength-trained athletes (84), and fascicle length of the gastrocnemius and vastus lateralis has been found to be greater in sprinters compared to distance runners (6). The combination of resistance, sprint, and jump training have been shown to increase rectus femoris fascicle length, and sprint and jump training have been shown to increase vastus lateralis fascicle length (24). These architectural changes have a positive effect on the manner in which force is ultimately transmitted to tendons and bones.

Other Muscular Adaptations

Resistance training has been shown to increase myofibrillar volume (107), cytoplasmic density (111), sarcoplasmic reticulum and T-tubule density (9), sodium–potassium ATPase activity (67), and increased ribosome biogenesis (54, 172). Collectively, these changes act to facilitate hypertrophy and enable greater expression of muscular strength. Sprint training has been shown to enhance calcium release (129), which assists in increasing speed and power production by promoting actin and myosin crossbridge formation.

Heavy resistance training has also been shown to reduce mitochondrial density (112). While the number of mitochondria remains constant or can slightly increase throughout a training phase, mitochondrial density is expressed relative to muscle area. Increases in muscle CSA occur disproportionately to mitochondrial proliferation, and consequently, the density of mitochondria per unit volume is typically found to decrease with hypertrophy. Muscle hypertrophy can result in decreased capillary density by similar mechanisms, again with the number of capillaries per fiber increasing somewhat (168, 184). Bodybuilding workouts produce large hydrogen ion concentrations, but having more capillaries per fiber may assist in the clearance of metabolites from exercising muscle (93, 164).

Anaerobic exercise results in substantial reductions in muscle and blood pH (32), with several mechanisms regulating a change in acid–base balance during exercise. With adaptations to consistent acute changes in pH during training (i.e., increased H^+ concentration), buffering capacity can improve. This increased capacity then allows an athlete to better tolerate the accumulation of H^+ within the working muscle, resulting in delayed fatigue and greater muscular endurance (152). By its nature, high-intensity interval training (e.g., sprints and cycling) performed above the lactate threshold has been shown to significantly increase buffering capacity by 16% to 38% (21, 152). Elsewhere, athletes competing in anaerobic team sports have been shown to have higher buffering

capacity than endurance athletes and untrained control subjects (52).

Within skeletal muscle, substrate content and enzyme activity represent additional areas of adaptation in response to anaerobic training. Most notably, when ATP and creatine phosphate (CP) concentrations are repeatedly exhausted following bouts of intermittent high-intensity muscular contraction, the storage capacity of these high-energy compounds is increased via a supercompensation effect. MacDougall and colleagues (113) reported a 28% increase in resting CP and an 18% increase in ATP concentrations following five months of resistance training (i.e., 3 to 5 sets of 8-10 repetitions with 2-min rest periods). In addition, it appears that bodybuilding-style programs such as this, which stress fast glycolysis, may also be a potent stimulus for the enhancement of glycogen content, which was found to increase up to 112% (113).

Connective Tissue Adaptations

Bones, tendons, ligaments, fascia, and cartilage are examples of connective tissue. Anaerobic exercise imparts mechanical forces that cause deformation of specific regions of the skeleton. These forces, created by muscular actions on the tendinous insertion into bone, can be bending, compressive, or torsional. In response to **mechanical loading**, **osteoblasts** migrate to the bone surface and begin bone modeling ([figure 5.4](#)). Osteoblasts manufacture and secrete proteins—primarily collagen molecules—that are deposited in the spaces between bone cells to increase strength. These proteins form the **bone matrix** and eventually become mineralized as calcium phosphate crystals (**hydroxyapatite**). New bone formation occurs predominantly on the outer surface of the bone (**periosteum**), increasing diameter and strength.

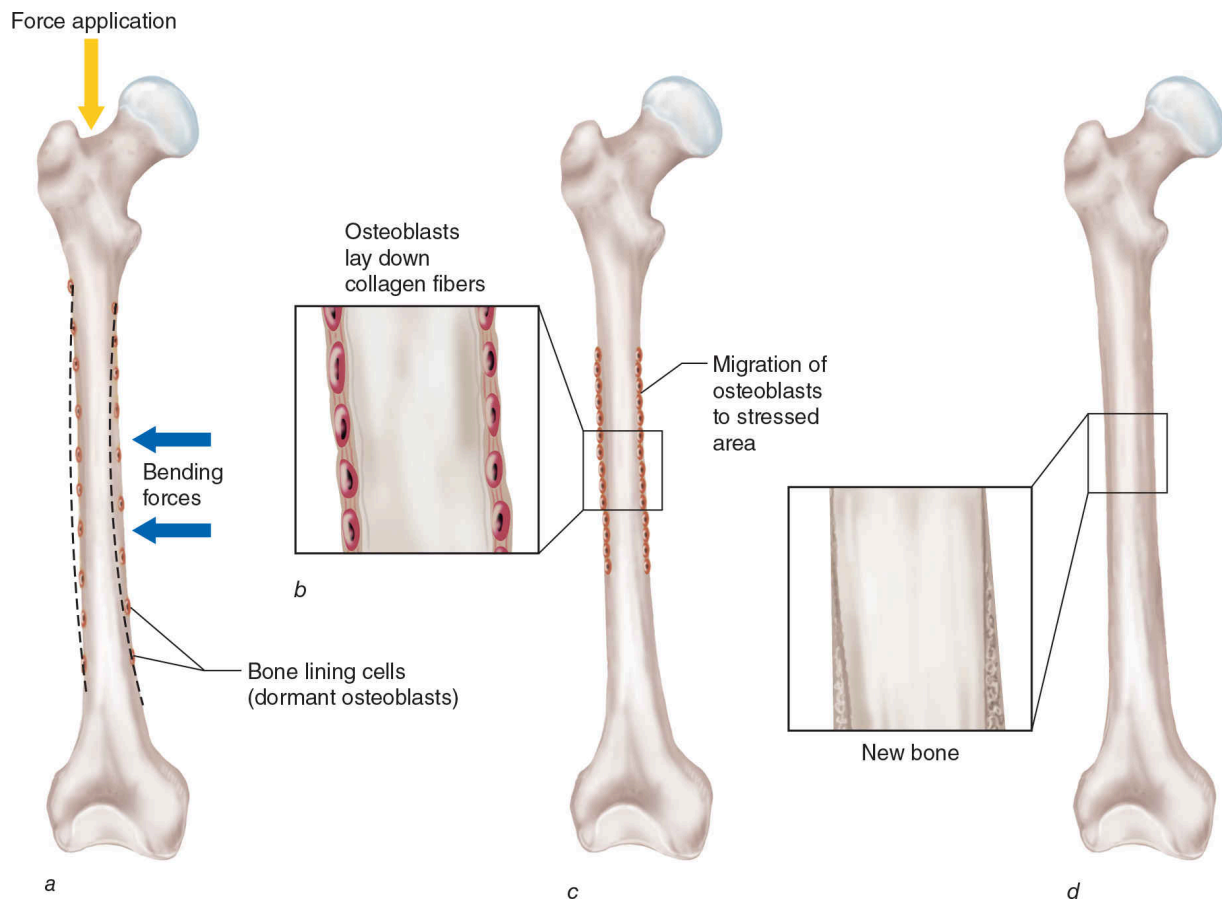


FIGURE 5.4 Bone modeling in response to mechanical loading. (a) Application of a longitudinal weight-bearing force causes the bone to bend (as depicted by the dotted line), creating a stimulus for new bone formation at the regions experiencing the greatest deformation. (b) Osteoblasts lay down additional collagen fibers at the site. (c) Previously dormant osteoblasts migrate to the area experiencing the strain. (d) The collagen fibers become mineralized, and the bone diameter effectively increases.

General Bone Physiology

The rate of bone adaptation occurs differently in the axial (skull-cranium, vertebral column, ribs, and sternum) and appendicular (shoulder girdle, pelvis, and bones of the upper and lower extremities) skeleton, owing to differing amounts of **trabecular** (spongy) **bone** and **cortical** (compact) **bone**. Cortical bone is dense and forms a compact outer shell surrounding the trabecular bone, with the two types of bone linked by interlocking narrow and delicate plates of trabecular bone. The spaces between the

trabecular plates are occupied by bone marrow that consists of adipose tissue and blood products such as immature red blood cells. Blood vessels from the marrow cavity extend into the dense cortical bone through a network of vertical and horizontal canals. Because it is less dense and has a greater surface area-to-mass ratio, trabecular bone is able to respond more rapidly to stimuli than cortical bone because it is softer, weaker, and more flexible and therefore more inclined to adaptive change.

The term **minimal essential strain (MES)** refers to the threshold stimulus that initiates new bone formation. Consistently exceeding these thresholds signals osteoblasts to migrate to the region experiencing the stress and to form bone, while forces that fall below the MES do not present an appropriate stimulus for new bone formation. Bone cells work to regulate the formation of new bone tissue such that forces experienced on a regular basis do not exceed the MES, thereby establishing a margin of safety against fracture. Strain registered by bone is a function of the force per unit area of bone (stress). The MES is thought to be approximately 1/10 of the force required to fracture bone. Increasing the diameter of bone allows the force to be distributed over a larger surface area, thereby decreasing the amount of mechanical stress. Following bone growth, a force that previously exceeded the MES will now be below the MES threshold. Progressive weight-bearing physical activities that generate forces exceeding the MES are therefore the most effective at increasing bone size and strength.

Forces that reach or exceed a threshold stimulus initiate new bone formation in the area experiencing the mechanical strain.

Anaerobic Training and Bone Growth

As muscular strength and hypertrophy increase in response to anaerobic training modalities, the forces generated by the increased muscle contractions subsequently increase the mechanical stress on

bone, and the bone itself must increase in mass and strength to provide an adequate support structure. Any increase in muscle strength or mass may therefore result in a corresponding increase in **bone mineral density (BMD)**, or the quantity of mineral deposited in a given area of the bone (93). Interestingly, inactivity or immobilization has the opposite effect and results in a more rapid rate of loss of bone matrix and BMD (155). Numerous studies have shown a positive correlation between BMD and muscle strength and mass (135, 174).

Researchers have reported that resistance-trained athletes have higher BMD than age-matched sedentary control subjects (12, 29). In some individuals (e.g., professional soccer players), physical activity seems to influence bone mass, area, and width more than BMD (174). Thus, exercise that stimulates muscle hypertrophy and strength gains also appears to stimulate bone growth. BMD is a key indicator of bone strength, but it does not encompass other crucial aspects of bone quality such as microarchitecture, geometry, and turnover. While **dual-energy X-ray absorptiometry (DXA)** is the most commonly used clinical tool for measuring BMD, it fails to assess these additional factors that contribute to overall bone quality. Consequently, there is increasing interest in employing other bone imaging techniques, such as **quantitative computed tomography (QCT)**, to evaluate bone microarchitecture (105). QCT is advantageous because it differentiates between cortical and trabecular bone compartments, highlighting their respective contributions to bone strength. For example, after a six-month resistance training program, older women showed significant increases in cortical bone density at the tibia and radius shaft regions using QCT, even though no significant changes were observed in DXA measurements (105). More research is needed to determine the best methods to increase bone strength, especially in young healthy adults.

Quantitatively, the time course for bone adaptations is rather long in most cases—approximately six months (28) or longer (102)—and

depends on the structure of the training program. For example, an 18-month resistance training program in middle-aged and older men led to significant increases in section modulus, a measure of bone strength, as well as CSA and areal BMD (aBMD) at the femoral neck, as assessed by QCT (102). However, adaptation begins within the first few workouts. The process of osteogenesis involves secretion of substances into the blood (substances specific to bone only) that can be measured. Therefore, any elevation in an osteogenic marker can be recognized as an early indicator of bone formation and presumably a precursor to increased BMD, provided that the stimulus is maintained over a long training period.

Principles of Training to Increase Bone Strength

Anaerobic training programs that have the objective to stimulate bone growth need to incorporate specificity of loading, speed and direction of loading, sufficient volume, appropriate exercise selection, progressive overload, and variation (30). **Specificity of loading** demands the use of exercises that directly load the particular region of interest of the skeleton. If the body interprets these forces as new or novel, they will stimulate bone growth in the area that is receiving the strain. For example, running may be a good stimulus for increased BMD in the femur but the wrong choice for promoting mineral deposits when one is trying to strengthen the wrist. The concept of specificity of loading becomes particularly important when a strength and conditioning professional prescribes exercises to increase bone mass in regions of the skeleton most commonly affected by **osteoporosis**—a disease in which BMD and bone mass become reduced to critically low levels. Research indicates that high-impact cyclical loading exercises for the lower body, such as gymnastics (161), volleyball, or basketball (47), selectively increase BMD at clinically relevant sites such as the hip and spine more than lower-impact activities do. Additionally, increases in BMD may be seen in highly trained college athletes

already possessing high levels of BMD. These changes in BMD are independent of reproductive hormonal status if the stimulus is sufficient (161).

Exercise selection is critical when one is trying to elicit maximal **osteogenic stimuli** (factors that stimulate new bone formation). In essence, exercises should involve multiple joints, should direct the force vectors primarily through the spine and hip (i.e., **structural exercises**), and should apply external loads heavier than those with single-joint assistance exercises. Cussler and colleagues (34) showed a positive linear relationship between the amounts of weight lifted over the course of a training year and associated increases in BMD. In addition, the findings of this research highlighted the importance of exercise specificity in that the squat, compared to seated leg press, was more effective for increasing BMD in the trochanter of the femur (34). The use of single-joint, machine-based exercises should be limited, because these exercises isolate a single muscle group by using equipment as support to stabilize the body rather than promoting skeletal support. Therefore, exercises such as the back squat, power clean, deadlift, snatch, and push jerk (for the axial skeleton and lower body) and the shoulder press (for the upper body) are recommended as more effective methods for increasing bone strength when indicated.

Because bone responds favorably to mechanical forces, the principle of **progressive overload**—progressively placing greater than normal demands on the exercising musculature—applies when one is training to increase bone mass (66, 170). Although the maximal strength of bone is maintained well above the voluntary force capabilities of the associated musculature, bone responds to higher forces (e.g., 1RM to 10RM loads) that are repetitively applied over time. The adaptive response of bone ensures that forces do not exceed a critical level that increases the risk of **stress fractures** (microfractures in bone due to structural fatigue). Support for progressive overload comes from studies that have compared BMD of various groups of athletes to that of nonathletes (47). In fact,

elite adolescent weightlifters have been found to possess levels of bone mineralization that far exceed values found in untrained adults (29, 81). This observation is interesting because it indicates that young bone may be more responsive to osteogenic stimuli than mature bone. Evidence indicates that physical activity during growth modulates the external geometry and trabecular architecture of bone, potentially enhancing skeletal strength (74). Evidence has shown that physical activity–associated bone loading during skeletal growth (i.e., adolescence) and after skeletal growth (i.e., early adulthood) elevates **peak bone mass** and is positively associated with adult bone mass in later life (83, 160).

Another important consideration in the design of programs to stimulate new bone formation is training variation. The internal architecture of the human skeleton has a mechanism through which it compensates for new strain patterns experienced by the bone. To optimally dissipate the imposed forces, the direction of the collagen fibers within the bone matrix may change to conform to the lines of stress experienced by the bone. Thus, changing the distribution (and direction) of the force vectors by using a variety of exercises continually presents a unique stimulus for new bone formation within a given region of bone. Subsequently, collagen formation occurs in multiple directions, increasing bone strength in various directions. Overall, if the magnitude of the load or the rate of force application is sufficient, it is not typically necessary to perform more than a total of 30 to 35 repetitions, because a greater volume of loading is not likely to provide any additional stimulus for bone growth (58, 69, 160).

The components of mechanical load that stimulate bone growth are the magnitude of the load (intensity), rate (speed) of loading, direction of the forces, and the volume of loading (number of repetitions).

How Can Athletes Stimulate Bone Formation?

- Select multijoint, compound exercises that involve many muscle groups at once.
- Select some exercises that direct axial force vectors through the spine and hip.
- Use both heavy-load exercises and ballistic or high-impact exercises to expose the bone to different intensities of force.
- Use multidirectional movements that incorporate all planes of motion to stimulate growth in different areas of the bone.
- Use the principle of progressive overload to stress the musculoskeletal system and continue to progressively increase load as the tissues become accustomed to the stimulus.

Adaptations of Tendons, Ligaments, and Fascia to Anaerobic Training

Tendons, ligaments, fascia, and cartilage are complex and dynamic structures that are the critical link between muscles and bones. The primary structural component of all connective tissue is the **collagen** fiber (type I for bone, tendon, and ligaments and type II for cartilage; [figure 5.5](#)). The parent protein, **procollagen**, is synthesized and secreted by fibroblasts, which are the most common cells found in the connective tissue of animals and act as stem cells in the synthesis of the extracellular matrix, as well as playing a critical role in wound healing.

Procollagen molecules consist of three protein strands twisted around each other in a triple helix. Procollagen leaves the cell with protective extensions on the ends to prevent premature collagen formation. Cleavage of the extensions via enzymes results in the formation of active collagen that aligns with other collagen molecules to form a long filament. Measurement of these enzymes provides an indication of collagen metabolism. In fact, enzyme levels increase in response to training, thereby showing increased net type

I collagen synthesis (103). The parallel arrangement of filaments is called a **microfibril**. Collagen has a striated (striped) appearance under a light microscope, somewhat like skeletal muscle, owing to the orderly alignment of the gaps between the collagen molecules within a microfibril. As bone grows, microfibrils become arranged into fibers, and the fibers into larger bundles. The true strength of collagen comes from the strong chemical bonds (**cross-linking**) that form between adjacent collagen molecules throughout the collagen bundles. Collagen bundles are bunched together longitudinally to form tendons or ligaments, or are arranged in sheets with the layers oriented in different directions, as found in bone, cartilage, and fascia.

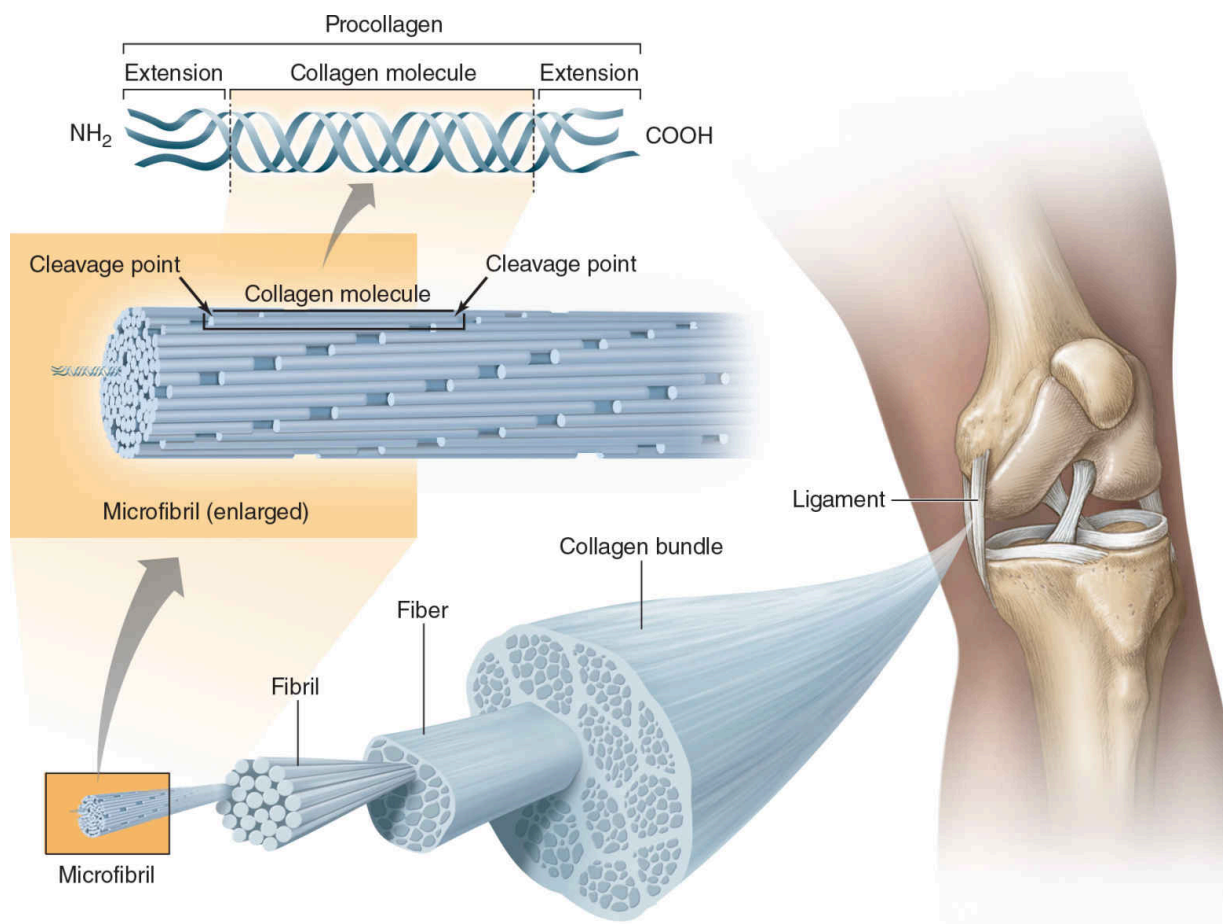


FIGURE 5.5 Formation of a collagen fiber.

Tendons and ligaments are composed primarily of tightly packed, parallel arrangements of collagen bundles. Mature tendons and ligaments contain relatively few cells. The small number of metabolically active cells in tendons and ligaments makes the requirement for oxygen and nutrients in these tissues relatively low. Ligaments contain elastic fibers (**elastin**) in addition to collagen, because a certain amount of stretch is needed within a ligament to allow normal joint motion. Tendons and ligaments attach to bone with great strength, allowing the maximal transmission of forces. The fibrous connective tissues that surround and separate the different organizational levels within skeletal muscle are referred to as *fascia*. Fascia has sheets of fibrocollagenous support tissue, containing bundles of collagen fibers arranged in different planes, to provide resistance to forces from different directions. Fascia within muscles converges near the end of the muscle to form a tendon through which the force of muscle contraction is transmitted to bone. Compared with that of muscle tissue, tendon metabolism is much slower due to poorer vascularity and circulation (82). In fact, the increase in blood flow to skeletal muscle via exercise is not paralleled by the same perfusion in tendons (88). This limited vascularity has implications for regeneration and is the reason tendons can take significant amounts of time to heal following injury.

The primary stimulus for growth of tendons, ligaments, and fascia is the insult from mechanical forces created during high-intensity exercise. The degree of tissue adaptation appears to be proportional to the intensity of exercise (88). Consistent anaerobic exercise that exceeds the threshold of strain has a positive effect on stimulating connective tissue changes (82).

Empirical evidence suggests that connective tissues must increase their functional capabilities in response to increased muscle strength and hypertrophy. The sites where connective tissues can increase strength and load-bearing capacity are

- the junctions between the tendon (and ligament) and bone surface,

- within the body of the tendon or ligament, and
- in the network of fascia within skeletal muscle (99).

As muscles become stronger, they pull on their bony attachments with greater force and cause an increase in bone mass at the tendon–bone junction and along the line over which the forces are distributed.

High-intensity anaerobic training results in connective tissue growth and other ultrastructural changes that enhance force transmission. Specific changes within a tendon that contribute to its increase in size and strength include the following:

- An increase in collagen fibril diameter
- A greater number of covalent cross-links within the hypertrophied fiber
- An increase in the number of collagen fibrils
- An increase in the packing density of collagen fibrils

Collectively, these adaptations increase the tendon's ability to withstand greater tensional forces (120).

Muscle hypertrophy in animals relates to an increase in the number and size of fibroblasts, thereby resulting in a greater supply of total collagen. Activation of fibroblasts and subsequent growth of the connective tissue network are prerequisites for hypertrophy of active muscle (121). This may explain why biopsies of trained athletes have shown that hypertrophied muscle contains greater total collagen than in untrained individuals but that the collagen content remains proportional to the existing muscle mass (120). Past studies indicate that **tendon stiffness** (force transmission per unit of strain, or tendon elongation) increases as a result of resistance training (101, 123). In fact, Kubo and colleagues (99) reported a 15% to 19% increase in Achilles tendon stiffness following eight weeks of resistance training. Furthermore, patella tendon CSA increased by 5% to 6% after nine weeks of knee extension training (151). The intensity of exercise is critical; heavy loads (80% 1RM) increase tendon stiffness, but light loads (20% 1RM) do not (100).

How Can Athletes Stimulate Connective Tissue Adaptations?

Tendons, Ligaments, Fascia

- Long-term adaptations in tendons, ligaments, and fascia are stimulated through progressive high-intensity loading patterns using external resistances.
- High-intensity loads should be used, because low to moderate intensities do not markedly change the collagen content of connective tissue.
- Forces should be exerted throughout the full range of motion of a joint, and wherever possible multijoint exercises should be used.

Cartilage

- Moderate-intensity anaerobic exercise seems to be adequate for increasing cartilage thickness. Strenuous exercise does not appear to cause any degenerative joint disease when progressively overloaded appropriately.
- Tissue viability can be maintained by adopting a variety of exercise modalities and ensuring that load is applied throughout the range of motion.

Adaptations of Cartilage to Anaerobic Training

Cartilage is a dense connective tissue capable of withstanding considerable force without damage to its structure. The main functions of cartilage are to

- provide a smooth joint articulating surface,
- act as a shock absorber for forces directed through the joint, and
- aid in the attachment of connective tissue to the skeleton.

A feature unique to cartilage is that it lacks its own blood supply and must depend on diffusion of oxygen and nutrients from synovial fluid (which is why cartilage does not easily repair itself following injury). Two primary types of cartilage are significant in relation to physical activity. **Hyaline cartilage** (articular cartilage) is found on the articulating surfaces of bones. **Fibrous cartilage** is a very

tough form of cartilage found in the intervertebral disks of the spine and at the junctions where tendons attach to bone.

The fact that articular cartilage gets its nutrient supply via diffusion from synovial fluid provides a link for joint mobility to joint health. Movement about a joint creates changes in pressure in the joint capsule that drive nutrients from the synovial fluid toward the articular cartilage of the joint (156). Immobilization of a joint prevents proper diffusion of oxygen and essential nutrients throughout the joint. This results in the death of the healthy cells within cartilage, called *chondrocytes*, and a resorption of the cartilage matrix (167). Current understanding indicates that human cartilage undergoes atrophy, or thinning, when external loading is removed (e.g., postoperative immobilization and paraplegia). However, the effect that increased external loading has on average cartilage thickness remains to be fully elucidated (50). In any case, it is likely that genetics plays a greater role in determining cartilage morphology.

Endocrine Responses and Adaptations to Anaerobic Training

Hormones have a variety of regulatory roles during anaerobic training and affect homeostatic mechanisms dedicated to keeping the body's functions within normal range during rest and exercise (60, 61, 102, 106, 109). These include the development of muscle, bone, and connective tissue through both anabolic and catabolic processes. As discussed in [chapter 4](#), endocrine responses to anaerobic training can include chronic changes in the acute response to a workout, chronic changes in resting concentrations, and changes in hormone–receptor content.

Chronic Changes in the Acute Hormonal Response

Adherence to a long-term resistance training program results in an increased ability to exert greater levels of muscular force (72), with relative training intensities increasing over time as the body adapts to tolerate progressively heavier loads. As a consequence, acute endocrine responses to anaerobic training will likely mirror these improvements, as has been predominantly shown with growth hormone (33). The longitudinal changes in endocrine function reflect the increased exercise stress being tolerated by the body in response to incremental external loading. It is thus hypothesized that any chronic adaptations in acute hormonal response patterns potentially augment the ability to better tolerate and sustain prolonged higher exercise intensities.

Chronic Changes in Resting Hormonal Concentrations

Chronic changes in resting hormone concentrations following anaerobic exercise are unlikely, with research providing inconclusive changes in testosterone, growth hormone, IGF-1, and cortisol over

time (96). Instead, resting concentrations likely reflect the current state of the muscle tissue in response to substantial changes to the training program (i.e., volume or intensity) and nutritional factors. It appears that the elevation during and immediately following a workout may present receptors with enough of a stimulus to affect tissue remodeling without the need for chronic elevations in basal concentrations (36). It is important to note that chronic elevations in an anabolic hormone may be counterproductive over the long term. Receptors tend to downregulate over time when exposed consistently to high levels of hormones. For example, in type 2 diabetes mellitus, sensitivity of skeletal muscle to insulin is reduced due to chronic elevated blood insulin. This is why anabolic steroid users repeatedly cycle drug use rather than maintaining consistently high doses.

Hormone Receptor Changes

Receptor content is important for mediating the adaptations elicited by any hormonal response. Androgen receptors (AR) have received much attention in the literature, and their content (i.e., number of receptors per area on the target tissue) depends on several factors including muscle fiber type, contractile activity, and the concentrations of testosterone. Resistance training has been shown to upregulate AR content within 48 to 72 hours after the workout (73). The resistance exercise stimulus appears to mediate the magnitude of acute AR modifications. Ratamess and colleagues (137) compared 1 set versus 6 sets of 10 repetitions of squats and reported no differences in AR content following the single-set protocol; however, the higher-volume protocol elicited significant downregulation of AR content 1 hour after the workout. This study also demonstrated that when sufficient volume is reached, AR protein content may initially downregulate before the upregulation that has been shown in other studies. However, Kraemer and colleagues (98) have shown that consumption of a protein–

carbohydrate supplement before and after the workout attenuates this AR downregulation.

Cardiovascular Adaptations to Anaerobic Training

Anaerobic exercise has important effects on the cardiovascular system that include both chronic adaptations at rest and acute adaptations (55-57). These changes include reductions in resting heart rate and blood pressure (55). Chronic resistance training also leads to increases in stroke volume and modifications in cardiac dimensions, such as increased left ventricular wall thickness and mass, although these changes are relative to body surface area and lean body mass (57). Additionally, resistance-trained athletes exhibit lower heart rates and blood pressure during acute exercise bouts compared to untrained individuals. Enhanced stroke volume and cardiac output are also noted, demonstrating improved efficiency in managing exercise-induced stress (55-57).

Chronic Cardiovascular Adaptations at Rest

The effect of anaerobic training modalities on resting heart rate remains to be fully elucidated. Short-term resistance training has been shown to decrease resting heart rate between 5% and 12% (55). However, when this effect is studied longitudinally over time, mixed responses are reported, with either no change in resting heart rate or reductions of 4% to 13% (53, 57). In chronically resistance-trained athletes (e.g., bodybuilders, powerlifters, weightlifters), both average and lower-than-average resting heart rates (60-78 beats/min) have been reported compared to those in untrained individuals.

A meta-analysis of resting blood pressure indicated that both systolic and diastolic blood pressure decreased by 2% to 4% as an adaptation to resistance training (85). It appears that the response is greater in those individuals who initially have elevated blood pressure. Similarly, the **rate-pressure product** (heart rate $\times$

systolic blood pressure; a measure of myocardial work) has been shown to either remain constant or decrease following resistance training. Stroke volume has been shown to increase in absolute magnitude but not relative to body surface area or lean body mass. That is, stroke volume will increase as lean tissue mass increases during long-term resistance training. Finally, resistance training may either not change or slightly decrease total cholesterol and low-density lipoproteins and increase high-density lipoproteins (80). Therefore, heavy resistance training does little to enhance resting cardiac function, but greater improvements may result from adaptations to a high-volume program with short rest periods (i.e., bodybuilding or circuit training) in which the overall continuity of the exercise stress in a workout is much higher.

Chronic resistance training also alters cardiac dimensions. Increased left ventricular wall thickness and mass have been reported, but the increase disappears when expressed relative to body surface area or lean body mass. It is thought that this increase may result from exposure to intermittently elevated blood pressures and increases in intrathoracic pressure in addition to accommodating changes from increases in lean body mass and body size. Highly resistance-trained athletes have greater than normal absolute posterior left ventricular and intraventricular septum wall thickness (57). Little or no change in left ventricular chamber size or volume is observed with resistance training; this is a major difference between resistance exercise and aerobic exercise. Greater than normal absolute left and right ventricular end-diastolic and end-systolic volumes have been reported in bodybuilders but not weightlifters (55), which indicates that high-volume training may be more conducive to increasing absolute left ventricular volumes. It is important to note that bodybuilders frequently incorporate aerobic exercise into their training programs in an effort to metabolize body fat and promote a lean body composition; therefore, it is possible that some of these adaptations have been brought about, in part, by aerobic endurance training. Bodybuilders as well as weightlifters

have greater than normal absolute and relative (to lean body mass and body surface) left atrial internal dimensions, with the bodybuilders showing a significantly greater dimension (46).

Chronic Adaptations of the Acute Cardiovascular Response to Anaerobic Exercise

Chronic resistance training reduces the cardiovascular response to an acute bout of resistance exercise of a given absolute intensity or workload. Short-term studies have shown that resistance training is associated with reductions in the acute increases in heart rate, blood pressure, and double product caused by submaximal resistance exercise but may promote an increase in the pressor response to maximal or isometric exercise (119, 142). In addition, male bodybuilders have been found to have lower systolic and diastolic blood pressure and heart rates during sets of 50% to 100% of 1RM performed to momentary muscular failure compared to both sedentary and lesser-trained men (56). Finally, oxygen extraction is generally not improved with resistance training using heavy loads and low volume. It is enhanced to a greater extent with continuous aerobic exercise or with a resistance training program using high volume and short rest periods or among populations with low baseline cardiorespiratory fitness (97, 119a).

Compatibility of Aerobic and Anaerobic Modes of Training

Strength or power training and work capacity–endurance training have divergent physiology that presents a programming challenge to the strength and conditioning professional working to optimize concurrent gains in both these physical and physiological characteristics. Combining resistance and aerobic endurance training may interfere with strength and power gains, primarily if the aerobic endurance training is high in intensity, volume, or frequency and compromises resistance training quality (51, 76, 106). Possible explanations for this less-than-optimal power development are likely due to the impact of aerobic training on the quality of resistance training via fatigue pathways, thus augmenting the resistance exercise stimulus (126). In contrast, most studies have shown no adverse effects on aerobic power resulting from heavy resistance exercise despite the expected cellular changes caused by this type of exercise (112). Few studies have shown that resistance training can hinder $\dot{V}O_2\text{max}$ improvements (62). Interestingly, Kraemer and colleagues (94) reported that women who performed both resistance exercise and aerobic endurance training had greater aerobic development than those who performed the aerobic endurance training alone. Such data have encouraged some athletes (e.g., distance runners) to add supplemental sport-specific resistance training to their total training regimen. Indeed, the majority of research indicates that heavy resistance training has very limited, if any, negative effects on aerobic power (126) but instead can serve to enhance performance in aerobic endurance sports (147, 169).

In a study examining the potential incompatibility of strength and endurance exercise, Kraemer and colleagues (94) used three months of simultaneous high-intensity strength and aerobic endurance training under five conditions:

1. A combination group (C) that performed both resistance and aerobic endurance training
2. A group (UC) that performed upper body resistance and aerobic endurance training
3. A resistance training-only group (S)
4. An aerobic endurance training-only group (E)
5. A control group

The S group increased 1RM strength and rate of strength development more than did the C group. Additionally, maximal oxygen consumption improvements were not affected by the simultaneous training (i.e., almost identical improvements in 2-mile [3.2 km] run times). Thus, no overtraining state for aerobic endurance was apparent.

A fascinating finding of the research by Kraemer and colleagues (94) was change in the muscle fiber size of the thigh musculature. Previous studies have shown decreases in muscle fiber size during aerobic endurance training (19, 62). Kraemer's group (94), however, indicated that the transformation of type IIx to type IIa fibers was almost complete in the S group ($19.1\% \pm 7.9\%$ type IIx before training to $1.9\% \pm 0.8\%$ after training) and C group ($14.11\% \pm 7.2\%$ before to $1.6\% \pm 0.8\%$ after). Interestingly, the UC and E groups (who performed only interval training) also had significant type IIx fiber transformation after training ($22.6\% \pm 4.9\%$ before to $11.6\% \pm 5.3\%$ after for UC, and $19.2\% \pm 3.6\%$ before to $8.8\% \pm 4.4\%$ after for E). This indicates that heavy resistance training recruits more type IIx fibers than high-intensity aerobic endurance interval training. Additionally, a small number ($<3\%$) of type IIa fibers were converted to type IIc fibers in the E group. The C group increased muscle size only in type IIa fibers, while the S group demonstrated increases in type I, IIc, and IIa fibers. The lack of change in type I fiber area and increase in type IIa fiber area in the C group appear to represent a cellular adaptation that shows the antagonism of simultaneous strength and aerobic endurance stimuli, since strength

training alone produced increases in both type I and II muscle fiber areas. The E group showed decreased size in type I and IIc fibers, presumably due to higher observed cortisol levels (and reduced testosterone) and their physiological need for shorter distances between capillaries and cells to enhance oxygen kinetics. On the other hand, other studies suggest that aerobic exercise can be done before resistance training without compromising performance (105a).

The majority of studies use untrained subjects to examine the effects of simultaneous high-intensity resistance training and aerobic endurance training (62, 104, 118). Few studies have looked at concurrent training in elite athlete populations (147). Several studies have shown an incompatibility using three days a week of resistance training alternating with three days a week of aerobic endurance training (i.e., training on six consecutive days), or four to six days a week of combined high-intensity resistance and aerobic endurance training (20, 48, 94), thereby lending credibility to the suggestion that the onset of overtraining mechanisms could have played a role. When both modalities are performed during the same workout (yielding a frequency of three days a week with at least one day off between workouts), the incompatibility has not been shown as frequently (104, 117, 118). An exception was a study by Sale and colleagues (141), which showed that training four days (two days of resistance training and two days of aerobic endurance training) per week was better than training two days (combined resistance training and aerobic endurance training) per week for increasing leg press 1RM (25% vs. 13%). These studies show that increasing the recovery period between workouts may decrease the incompatibility, a principle supported by the literature (173, 126).

What Performance Improvements Occur Following Anaerobic Exercise?

Muscular Strength

- A review of more than 100 studies has shown that mean strength can increase approximately 40% in untrained, 20% in moderately trained, 16% in trained, 10% in advanced, and 2% in elite participants over periods ranging from four weeks to two years (10).
- With training, a positive shift in muscle fiber types reflects an augmented recruitment of higher-order motor units. Type IIx fibers transition to type IIa fibers and reflect a greater fatigue resistance at similar absolute force output as well as an increase in muscle myofibrillar content through hypertrophy (146).

Power

- The optimal load for maximizing absolute peak power output in the jump squat is 0% of the 1RM (i.e., body weight [31]). However, it has been reported that peak power output is maximized in trained power athletes with higher loads corresponding to 30% to 60% of a squat 1RM (16).
- Peak power in the squat is maximized at 56% of the 1RM and in the power clean at 80% of the 1RM (31).
- For the upper body, peak power output can be maximized during the ballistic bench press throw using loads corresponding to 46% to 62% of a bench press 1RM (15, 128).

Local Muscular Endurance

- Cross-sectional data in anaerobic athletes have shown enhanced muscular endurance and subsequent muscular adaptations consistent with improved oxidative and buffering capacity (60).
- Skeletal muscle adaptations to anaerobic muscular endurance training include fiber-type transitions from type IIx to type IIb, as well as increases in mitochondrial and capillary numbers, buffering capacity, resistance to fatigue, and metabolic enzyme activity (60).

Body Composition

- Resistance training can increase fat-free mass and reduce body fat by up to 9% (116).
- Increases in lean tissue mass, daily metabolic rate, and energy expenditure during exercise are outcomes of resistance training (53).

Flexibility and Mobility

- Anaerobic training potentially can have a positive impact on flexibility, and a combination of resistance training and stretching appears to be the most effective method to improve flexibility with increasing muscle mass (8). When exercises are performed properly through a full range of motion, mobility may

be enhanced or maintained through resistance training with loads greater than only body mass.

Aerobic Capacity

- In untrained people, heavy resistance training can increase $\dot{V}O_{2\max}$ from 5% to 8%. In trained individuals, resistance training does not significantly affect aerobic capacity (52).
- Circuit training, high-intensity interval training, and programs using high volume, high intensities, and short rest periods (i.e., 30 s or less) have been shown to improve $\dot{V}O_{2\max}$ (25).

Motor Performance

- Resistance training has been shown to increase running economy, vertical and broad jumps, sprint speed, cycling power, tennis serve velocity, swinging and throwing velocity, and kicking performance (116).

Power development appears to be negatively affected more than strength during concurrent high-intensity resistance and aerobic endurance training. Häkkinen and colleagues (70) reported similar dynamic and isometric strength increases following 21 weeks of concurrent training or resistance training only; however, the resistance training-only group showed improvements in the rate of force development whereas the concurrent training group did not match this increase. Kraemer and colleagues (94) also showed that a resistance training-only group increased muscle power whereas a combined group did not. The resistance training-only group also increased peak power in upper and lower body tests while the combined group did not. It appears that power development is much more susceptible to the antagonistic effects of combined strength and aerobic endurance training than slow-velocity strength (94). Finally, the sequence may play a role in the magnitude of adaptation (51).

Detraining

Detraining is the term given to a decrement in performance and loss of the accumulated physiological adaptations following the cessation of anaerobic training or when there is a substantial reduction in frequency, volume, intensity, or any combination of these variables. According to the principles of *reversibility*, training-induced adaptations are transient and thus can disappear when the training load is insufficient or removed completely. The result is a partial or complete loss of the anatomical, physiological, and performance adaptations that have been made (123). The magnitude of these losses is dependent on the length of the detraining period as well as the initial training status of the individual.

Following the removal of a training stimulus is a lag time before the effects of detraining are fully observed. Strength performance in general is readily maintained for up to four weeks of inactivity, but in highly trained athletes, eccentric force and sport-specific power may decline significantly faster (124). In trained weightlifters, removing the training stimulus for 14 days did not significantly affect 1RM strength performance in the bench press (-1.7%) and squat (-0.9%), isometric (-7%) and isokinetic concentric knee extension force (-2.3%), or vertical jump performance (1.2%) (108). In recreationally trained men, very little change is seen during the first six weeks of detraining (92). Longer periods of training cessation are accompanied by significantly pronounced declines in the strength performance of strength-trained athletes (124), but this loss is still limited to 7% to 12% during periods of inactivity ranging from 8 to 12 weeks.

This strength loss is coupled with decreased average maximal bilateral and unilateral intramuscular EMG. Indeed, strength reductions appear related to neural mechanisms initially, with atrophy predominating as the detraining period extends. Interestingly, the amount of muscle strength retained is rarely lower

than pretraining values, indicating that resistance training has a residual effect when the stimulus is removed. However, when the athlete returns to training, the rate of strength reattainment is high, supporting the paradigm of muscle memory (153).

With respect to muscle fiber characteristics following the cessation of training, it appears that fiber disruption remains unchanged during the initial weeks of inactivity, but oxidative fibers may increase in strength-trained athletes (decrease in endurance athletes) within eight weeks of stopping training (124). In strength-trained athletes, 14 days of inactivity has been reported to have no effect on muscle fiber type distribution (78). In comparison, muscle fiber CSA declines rapidly in strength and sprint athletes (124). In 12 trained weightlifters, Hortobagyi and colleagues (78) observed a decline of 6.4% in fast-twitch fiber CSA in 14 days. These changes specifically targeted fast-twitch fibers initially, with no significant change immediately found in the slow-twitch fiber population. Longer periods of stoppage bring about declines in both fast- and slow-twitch fiber CSA and muscle mass in anaerobically trained athletes.

Conclusion

Anaerobic exercise represents a specific type of training stress to the anatomy and various systems of the body, and adaptations consequent to anaerobic training are specific to the nature of the exercise performed. It is also apparent that an individual's age, nutrition, prior fitness level, and training motivation can affect adaptations. Integration of a training program that has many components requires careful planning and monitoring to minimize the occurrence of incompatibilities or overtraining. Explosive training evokes marked increases in muscular power, whereas more conventional heavy resistance training mainly increases muscle size and strength. Anaerobic training (resistance, sprint, plyometric, agility, high-intensity interval training) in general elicits specific adaptations in the nervous system leading to greater recruitment, firing rate (rate coding), and enhanced muscle function that increases strength and power.

Anaerobic training also has positive effects on bone, muscle, and the associated connective tissue; the entire musculoskeletal system undergoes a coordinated adaptation to exercise. Athletes undertaking strenuous exercise training experience changes in the force-generating capabilities of muscle, resulting in a coordinated and proportional increase in the load-bearing capacity of bone and other connective tissue. Anaerobic training can increase skeletal muscle mass, force-generating capability, and metabolic capacity and may lead to subtle alterations of the endocrine system that enhance the tissue remodeling process.

Anaerobic training generally results in fewer acute and chronic responses in the cardiovascular and respiratory systems, although low-intensity, high-volume resistance exercise produces some responses that are similar to those with aerobic exercise. Collectively, improved neuromuscular, musculoskeletal, endocrine, and cardiovascular function contribute to enhanced muscle strength, power, hypertrophy, muscular endurance, and motor performance—

all of which contribute to increased athletic performance. The adaptations observed in athletes are directly related to the quality of the exercise stimulus and accordingly to the levels of progressive overload, specificity, and variation incorporated into program design. The scientific basis of program design is ultimately seen in the effectiveness with which the athlete improves performance.

KEY TERMS

actin
anaerobic training
bilateral deficit
bilateral facilitation
bone matrix
bone mineral density (BMD)
collagen
cortical bone
cross-education
cross-linking
cross-sectional area (CSA)
detraining
dual-energy X-ray absorptiometry (DXA)
elastin
electromyography (EMG)
exercise-induced muscle damage (EIMD)
fibrous cartilage
hyaline cartilage
hydroxyapatite
hyperplasia
hypertrophy
mechanical loading
microfibril
minimal essential strain (MES)
motor unit
myogenesis

myosin
myotatic reflex
nebulin
neuromuscular junction (NMJ)
osteoblasts
osteogenic stimuli
osteoporosis
peak bone mass
pennation angle
periosteum
procollagen
progressive overload
quantitative computed tomography (QCT)
rate–pressure product
size principle
specificity of loading
stress fractures
structural exercises
tendon stiffness
titin
trabecular bone

STUDY QUESTIONS

1. Which neuromuscular phenomenon occurs when there is an increase or maintenance of muscular strength of an untrained limb when training the contralateral limb?
 - a. cross-education
 - b. bilateral deficit
 - c. exercise-induced muscle damage
 - d. maximal voluntary isometric contraction
2. Adaptations to anaerobic training, particularly neuromuscular system improvements that occur during the early phases of training, can occur in

- a. 1 to 4 weeks
 - b. 6 to 8 weeks
 - c. 10 to 14 weeks
 - d. 20 to 24 weeks
3. Which of the following is not one of the motor unit adaptations to anaerobic training?
- a. hypertrophy of all types of muscle fibers
 - b. greater synaptic input needed for re-recruitment
 - c. shift of neuron firing frequency to higher rates
 - d. recruitment of a maximum percentage of available motor units
4. Bone development during anaerobic training requires a minimal essential strain (MES) to trigger the appropriate adaptations. What is the MES required?
- a. 1/2 the fracture force
 - b. 1/5 the fracture force
 - c. 1/10 the fracture force
 - d. 1/15 the fracture force
5. Anaerobic training results in what muscle fibrillar adaptations?
- a. Fibers, as a whole, shift to become more anaerobic.
 - b. Fibers, as a whole, exhibit no change.
 - c. Fibers, as a whole, shift to become more aerobic.
 - d. Anaerobic fibers become more anaerobic, while no change is observed in aerobic fibers.